

Exercise: Machine Learning User Segmentation

Your task is to analyze a dataset of newly created user accounts (all created within the past year) and develop a machine-learning segmentation model that groups users based on their early behaviors, demographics, and contribution patterns.

The goal is not to produce a specific “correct” set of clusters. Instead, we want to understand how you think, how you frame ambiguous analytical problems, how you engineer meaningful features, and how you communicate insights to non-technical stakeholders. We are especially interested in your ability to uncover distinct user segments and interpret what makes the groups meaningfully different from each other.

Deliverables

Workflow

Before beginning your analysis, provide a short description of how you plan to structure your work, including:

- How you interpret the goal and constraints
- What questions you will answer first and why
- What your analysis workflow will look like
- What assumptions you need to make
- What risks, edge cases, or data quality issues you anticipate
- How you will make the analysis reproducible (folder structure, code modularity, versioning, etc.)

Exploratory Data Analysis

Conduct an exploratory data analysis that examines:

- Summary statistics and distributions of key engagement metrics
- Activity patterns (e.g., days logging in vs. names added)
- Demographic differences
- Missing data and potential biases
- Relationship patterns between early engagement and contribution patterns

Modeling

Build one or more machine-learning models to identify meaningful user segments. You may use whichever technique you’d like —please explain your choice.

Include:

- Rationale for your chosen segmentation approach
- Feature selection and feature engineering
- How you determined or evaluated the number of segments
- Methods used to validate or interpret cluster quality
- Clear descriptions of the resulting segments and how they differ

We are more interested in your reasoning and interpretation than in algorithmic optimization.

Insights and Recommendations

Translate your findings into clear, business-oriented insights and present the findings in whatever format you choose.

Your insights might include:

- What distinguishes the segments from each other
- Which segments represent high-engagement or high-contribution potential
- How early user behavior patterns differ across the groups
- Recommendations for targeted onboarding, communication, or product improvements
- Any surprising patterns or opportunities you discovered

Assume your audience includes senior leaders—keep it concise and interpretable.